

Exercise 1 - Quantum Computation

Please upload your solutions to the moodle system. Submissions are individual. We expect fully detailed answers. If something is unclear about the exercise, mail caspero@post.bgu.ac.il. The deadline is 26/4/2026 at midnight.

1 Inner Products, Angles and Measurement of a single qubit.

Let $|\psi\rangle = \cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle$.

- a) Write the state $|\psi\rangle$ using the standard notation. Write $\langle\psi|$ using the standard notation.
- b) Verify that $||\psi\rangle||_2 \equiv \sqrt{\langle\psi|\psi\rangle} = 1$.
- c) What is the inner product between $|\psi\rangle$ and $|0\rangle$? What is the angle between $|\psi\rangle$ and $|0\rangle$? (Note: the angle between two vectors u, v is defined as $\arccos(\frac{\langle u, v \rangle}{||u|| \cdot ||v||})$)
- d) You are given a quantum state which is either $|\psi\rangle$ or $|0\rangle$ uniformly at random. You win if you are able to identify correctly the state that you are given, and you lose otherwise. Calculate the winning probabilities of the following strategies¹:
 - (a) Measure the qubit in the computational basis. Answer '0' if the outcome is '0', and answer ' ψ ' if the outcome is 1.
 - (b) Apply the Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and then measure the qubit in the computational basis.² Answer ' ψ ' if the outcome is '0', and answer '0' if the outcome is '1'.

¹These strategies are sub-optimal. You may try to find an optimal strategy - it isn't too hard.

²We will use the shorthand "measure in the Hadamard basis" or the H basis for this procedure in the future.

2 Tensor Product State

Let $|\alpha\rangle = a|0\rangle + b|1\rangle$, $|\beta\rangle = c|0\rangle + d|1\rangle$, where $a, b, c, d \in \mathbb{C}$. (Notice that this vector may not be normalized).

- a) Write the state $|\alpha\rangle \otimes |\beta\rangle$ in the $\{|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle\}$ basis, using both the Dirac notation and the usual matrix notation (i.e. as a column vector).
- b) Calculate the standard inner product between $|\alpha\rangle$ and $|\beta\rangle$.
- c) Calculate the norm of $|\alpha\rangle$, i.e. $\| |\alpha\rangle \|_2 = \sqrt{\langle \alpha | \alpha \rangle}$. Prove that

$$\| |\alpha\rangle \otimes |\beta\rangle \|_2 = \| |\alpha\rangle \|_2 \cdot \| |\beta\rangle \|_2.$$

- d) Prove or disprove: Is the state $|\psi\rangle \equiv \sqrt{\frac{1}{3}}|0\rangle \otimes |1\rangle + \sqrt{\frac{2}{3}}|1\rangle \otimes |0\rangle \equiv \sqrt{\frac{1}{3}}|01\rangle + \sqrt{\frac{2}{3}}|10\rangle$ a tensor product state?
- e) Same question as the previous item, but for the state $\sqrt{\frac{1}{3}}|11\rangle + \sqrt{\frac{2}{3}}|10\rangle$

3 Operations and Measurements of Multiple Qubits

Let $|\psi\rangle = \sqrt{\frac{1}{6}}(|00\rangle - 2|10\rangle + |11\rangle)$.

- a) Write the state $|\psi\rangle$ in the standard notation (i.e., as a column vector).
- b) We measure both the qubits of the state $|\psi\rangle$ in the $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ basis. What are the various probabilities of the result?
- c) We measure only the right qubit of the state $|\psi\rangle$. What are the probabilities of the outcomes 0 and 1? What is the state of the two qubits after the measurement, given that we measured “0”? Write the result also in the standard notation.
- d) What happens to the state if we first apply a Hadamard gate on the left qubit and only then measure the right qubit? Answer the same questions as in section (c).

4 Transformations of Bell's basis

In the lectures, we defined four states:

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)$$

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle)$$

$$|\phi^-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

These states are often called Bell's basis.

Suppose Alice holds the left qubit of $|\psi^+\rangle$ and Bob holds the other qubit.

- a) Show how Alice can transform $|\psi^+\rangle$ to the three other states, by acting only on her qubit.
- b) We defined the goal of super-dense coding in the last lecture before passover. Explain how to do super-dense coding. In more detail, Alice and Bob share the $|\psi^+\rangle$. Alice has a two-bit message $m = (m_1, m_2)$. Alice is allowed to use their entangled state and send one qubit to Bob. Bob needs to recover m . Design such a protocol.

(Remark: Using resource theories terminology, $q + e \rightarrow 2c$ where q represents the resource of sending a single qubit, e represents the resource of sharing an entangled pair, and c represents the resource of sending a classical bit. What other interesting relations of this type are satisfied? Teleportation is one such example, which we will soon see!)